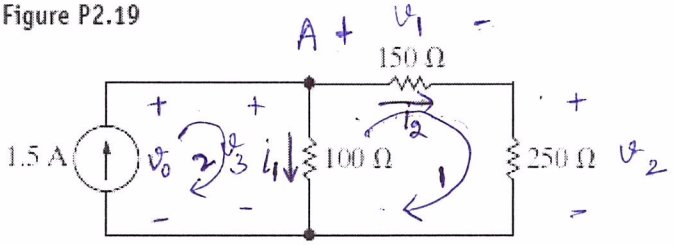
2.19 a) Find the currents i_1 and i_2 in the circuit in Fig. P2.19.

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b) Find the voltage v_o.

c) Verify that the total power developed equals the total power dissipated.

Figure P2.19



a) Apply KCL at node A

$$-1.5 + i_1 + i_2 = 0$$

$$\Rightarrow \boxed{i_1 + i_2 = 1.5} \quad (1)$$

Apply KVL to loop 1

$$v_1 + v_2 - v_3 = 0$$

$$i_2(150) + i_2(250) - i_1(100) = 0.$$

$$\boxed{4i_2 - i_1 = 0} \quad (2)$$

$$\boxed{i_1 = 1.2\text{A}, \; i_2 = 0.3\text{A}}$$

b) $v_3 - v_o = 0.$

$$v_o = v_3$$

$$= i_1(100)$$

$$= \underline{120\text{ V}}$$

c) $P_{developed} = (1.5)v_o$

$$= 1.5 \times 120 = 180\text{W}$$

$P_{dissipated} = i_1^2(100) + i_2^2(150) + i_2^2(250)$

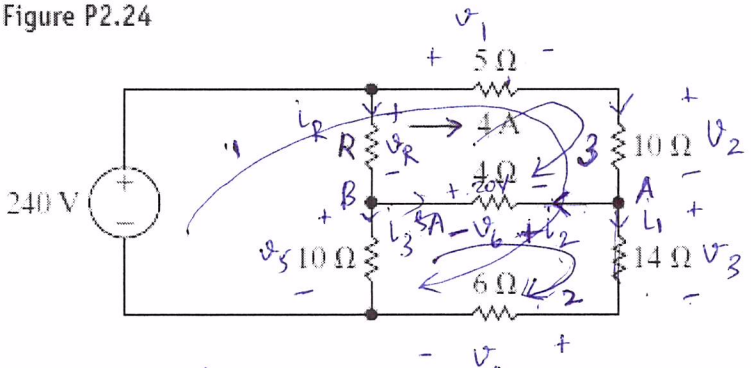
$= 180\text{W}$

2.24 For the circuit shown in Fig. P2.24, find (a) R and (b) the power supplied by the 240 V source.

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Figure P2.24



① Apply KVL to Loop 1

$$-240 + v_1 + v_2 + v_3 + v_4 = 0$$

$$-240 + 20 + 40 + i_1(14) + i_1(6) = 0$$

$$\boxed{i_1 = 9A}$$

② Apply KCL at node A

$$-4 + i_1 + i_2 = 0$$

$$\boxed{i_2 = -5A}$$

③ Apply KVL to loop 2

$$-v_6 + v_3 + v_4 - v_5 = 0$$

$$-i_2(4) + i_1(14) + i_1(6) - i_3(10) = 0$$

$$-(-5)(4) = +20 \qquad \boxed{i_3 = 20A}$$

④ Apply KCL at node B

$$-i_2 + i_3 - i_R = 0$$

$$i_R = i_3 - i_2 = 20 - (-5)$$

$$\boxed{i_R = 25A}$$

⑤ Apply KVL to loop 3

$$v_1 + v_3 + v_6 - v_R = 0$$

$$v_R = v_1 + v_3 + v_6$$

$$= 20 + 40 + 4(i_2)$$

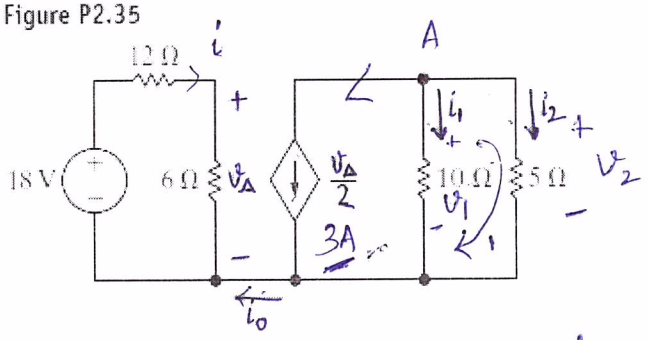
$$= 40V$$

⑥ Find R

$$R = \frac{v_R}{i_R} = \frac{40}{25} = 1.6\,\Omega$$

2.35 Find (a) i_o, (b) i_1, and (c) i_2 in the circuit in Fig. P2.35.

Figure P2.35



a) $i_o = 0$ as current cannot flow through a single conductor connecting 2 parts of a circuit.

b) $i = \frac{18}{(6+12)} = 1A$, $v_\Delta = i(6) = 6V$

$-3 - i_1 - i_2 = 0$

$i_1 + i_2 + 3 = 0$

$$\boxed{i_1 + i_2 = -3}$$

Applying KVL to loop 1

$v_2 - v_1 = 0$

$$\boxed{i_2(5) - i_1(10) = 0}$$

$$\boxed{\begin{aligned} i_1 &= -1A \\ i_2 &= -2A \end{aligned}}$$